Unit One: Critical Thinking is Fundamental

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ou have felt the results but probably did not know it had a name—around every 18 months a new generation of computers comes to market making yours even more obsolete. This phenomenon is known as Moore's Law. Gordon Moore, Intel co-founder, noted decades ago that the number of transistors contained on a microchip doubles around every 18 months. In 1971, 4004 transistors could be placed on one-square inch of a microprocessor; by 2012 that number had increased to 2,600,000,000.



Many scientists predict this doubling trend will end within the next few years. After all, although transistors continue to grow smaller, microchips are simply running out of room. If these scientists are correct, advancement in technology will dramatically slow and perhaps totally stagnate.

However, a handful of pioneering physicists challenge this assumption saying, "Fat chance!" They point to a new type of transistor made from a single phosphorus atom. Now, they claim, there is space for nearly an infinite number of transistors. Make ready, they predict, for an atomic, quantum computer that will appear *magical* compared to the conventional computers of today.

If these physicists are correct, their creation will spawn new technologies beyond anyone's ability to comprehend. We all will be profoundly transformed socially, economically, and environmentally. Nothing will be routine. Change will continue to accelerate; the world and our individual lives will become increasingly complex. A new, post-industrial world order may be at hand, ushering in new conveniences, opportunities, and lifestyles. But with it will undoubtedly come unimagined, multidimensional challenges and problems to unravel.

Change will begin on a personal level. Children and students will find it difficult to prepare for the future—much of what they learn in school will probably be obsolete when they begin their careers. Before the age of computers, most individuals began a vocation, stayed with it for thirty-five years or so, and retired around 65 years with a gold watch. However, today in a world that is continually shrinking, where political and economic realities increasingly overlap, students graduating from college can expect to change careers between four and seven times.

If all of this is true, how can a young person prepare to survive in a world where the rules are constantly changing, where change becomes the only constant, where one's success will be measured by one's adaptability? Workers will increasingly be called upon to become decision makers and exercise sound judgment. *Critical Thinking/Critical Writing* is dedicated to addressing these propositions. We believe that although no one can predict the future, critical thinking and writing skills are indispensable for confronting and adapting head-on with the future. The past will no longer guide us toward the future. Yes, the challenges are going to be immense—are you ready to meet them?

Chapter 1

Why Critical Thinking?

At the conclusion of this chapter, you will be able to

- Define critical thinking in your own words;
- Summarize the importance of critical thinking;
- Explain how writing is a natural extension of thinking;
- Justify your decision making process in a given situation;
- Apply Socratic inquiry to critical thinking.

Physicist Dr. Alexander Calandra relates an amusing, first-person narrative about a young student who protested his failing grade on his physics exam. The student claimed that he answered the instructor's exam question perfectly and deserved full credit while his instructor maintained the student's answer had nothing to do with physics. Unable to agree upon the student's grade, both the instructor and the student concurred they would arbitrate their differences using an impartial judge, Dr. Calandra.

The exam question stated: "*Show how it is possible to determine the height of a tall building with the aid of a barometer*." After studying the question momentarily, Dr. Calandra called the student into his office to hear the student's explanation.



Source: Snopes.com

The student responded that he wrote on the exam: "Tie a rope to the end of the barometer, take it to the top of the building and lower it to the ground. Then measure the length of the rope. That will tell you the height of the building."

Dr. Calandra agreed that using this technique would, indeed, determine the height of the building and provide a correct answer. However, Dr. Calandra was in a quandary to determine if the student should receive any credit for his answer. After all, the exam counted a significant portion of the student's final grade and would, in part, certify the student's understanding of basic physics. So, Dr. Calandra told the student that if he could provide another way of using the barometer to determine the building's height, he might recommend giving him credit. Dr. Calandra turned to the student, told him he had only a few minutes before he had to leave for his class, and then waited for the student's response.

After several minutes had gone by with no answer, Dr. Calandra asked the student if he just wanted to give up.

"Oh no," said the student. "There are many ways of getting the height of a tall building with the aid of a barometer. One way is to take the barometer to the top of the building, lean over, drop the barometer, and use a stopwatch to time its fall to the ground. Of course this will do irreparable damage to the barometer. But, using the formula H=0.5g x t squared you would have the height of your building.

"Or, on a sunny day, you could measure the height of the barometer and its shadow on the ground, and by using proportions, calculate the height of the building. Or, you could measure the height of the barometer and mark its length on the walls of the building while climbing the stairs. Then, simply add up all the lengths. Or, the easiest method would be to go down to the basement and offer to give the barometer to the building superintendent if he would tell me how high his building was."

Finally, needing to go to class and a bit exasperated, Dr. Calandra asked the student if he really knew the answer to the exam question. The student admitted that he knew all along the answer his instructor wanted. It simply involved taking two barometric readings, one from the top of the building and the other from the bottom, and by calculating the difference, he could determine the building's height. But, the student further explained, I am tired of professors requiring students to always memorize facts and formulas that they want us to regurgitate on their exams. We need to be challenged. Rather than being taught *what to think*, we should be taught *how to think*.

Today, no one is sure if Dr. Calandra's story is true or he merely intended it as a parable to illustrate a point. Many versions of this tale continue to be circulated. One version making the rounds in Denmark maintains that the precocious student was Nobel Prize physicists Neils Bohr who later worked on the Manhattan Project to create the first atomic bomb.

Just What is Critical Thinking?

Regardless of whether Dr. Calandra's story is true or merely folklore, his point is clear: the frustrated student insisted on approaching the barometer problem from his own perspective. He wanted to learn how to think critically and creatively to resolve problems, not merely "plug and play" what he had memorized from class. Although his professor may have been a bit irritated with the student's scheme for going about it, the student's desire to examine the problem from various angles suggests important questions for all of us to consider:

- How do we make sense of the world around us?
- How do we make reasoned judgments about anything?
- When we make a decision, how do we evaluate its potential consequences?
- Is there a connection between all types of learning?
- Are there skills I can learn to help me improve the quality of my thinking?
- How does writing reveal the quality of my thinking?
- How can I move beyond memorization to applying that knowledge creatively?
- Can I use college as an opportunity to find the answer to these questions?

Asking the Right Questions

Essentially, the remainder of this text is devoted to exploring these questions with you. We have designed this exploration to help you develop skills for improving your thought processes and ultimately realize the connection between your disciplines and all learning through the process of critical thinking and critical writing. Furthermore, as you work your way through this text, you will come to appreciate that language is an essential component of thought. Whether spoken or written, language helps us shape our ideas to better comprehend the world around us.



Illustration from the Renaissance of how the mind processes thought. Robert Fludd (1564-1637).

We will begin our investigation, then, by asking you to first *think about thinking*. That is, what actually constitutes thinking? Are there various levels of thought, and if so, what are the differences among those levels? Is there "good" thinking and "bad" thinking? If so, how does anyone recognize the difference? Finally, how can you improve the quality of your own thinking and by extension, your own writing?

Obviously, daydreaming in chemistry class about what you plan to do over spring break is considerably different from working through problems on covalence bonding and atomic orbitals. Random, stream-of consciousness thoughts about spring break at the beach certainly seem antithetical to pouring over a chemistry text, puzzling over the theories and formulas behind electron bonding. Yet, on further analysis, you will agree that planning a spring break trip, like working the chemistry problems, requires considerable thought, reasoned judgments, and good decision making if you are going to successful. Both require what we call *critical thinking*. Thinking, left unto itself, may be random and unproductive. Critical thinking, though, is disciplined thought where we actually intervene and acquire knowledge through cognition, "the mental process of acquiring knowledge by the use of reasoning, intuition, or perception."

You may feel this is a good opportunity to formally define *critical thinking*. But remember, this text is about critical thinking. What would our barometer student think if we asked you to memorize a definition without first asking you to think on your own about it? So first, begin thinking about this proposition yourself; that is, to *think about thinking*, and ultimately create your

own definition. Perhaps you already have a few ideas about what constitutes critical thinking. For the present, though, begin with a working definition by describing what you believe people do when they are critically thinking. Yes, we will offer a few definitions later in this chapter for you to compare your definition, but for the time being, *think about thinking*.

To begin, consider the following radio advertisement from a California car dealership. Amazingly, this ad proved very effective and increased the dealership's sales significantly.



Hey, folks! Need cash for Christmas? Maxed out your credit cards? More presents to buy than cash in your wallet? WE HAVE A SOLUTION! This week only, Bendover Car Dealership is going to give you $700 cash to go the mall of your choice. Yes, any person buying a new or used car this week will receive a $700 voucher to go shopping. And the good news you won't have to make any car payments until next March. Why not celebrate the holidays by buying nice gifts for your loved ones while cruising around in your sporty new car? We're open late until the last customer is served so come in this week to Bendover Car Dealership and get your free money!

Of course you are thinking no one would actually fall for this radio ad, but surprisingly, many people needing cash for the holidays actually bought cars to get their "free" money. Apparently, the enticement of a few quick bucks short-circuited these individuals' executive decision-making processes. We can only wonder if these people have ever considered why they are already drowning in debt, or how they can possibly take on more debt. Do you think they justified their decision for immediate gratification—live for the moment and not worry about future consequences?

Thinking involves making decisions; critical thinking necessitates considering the *consequences* of those decisions. Pause now to evaluate one major decision you have already made in life—perhaps to work, attend college, join the military, get married, purchase a home, have children, and so on. What factors did you weigh before making that decision? Was your decision logical, emotional, or a combination of both? How long did you plan before making the decision? Did you consider the long-range consequences of that decision? Finally, how did making that decision turn out for you?

Critical thinking, though, is more than making reasoned judgments and predicting the consequences. Critical thinking involves being able to solve problems as well. For example,

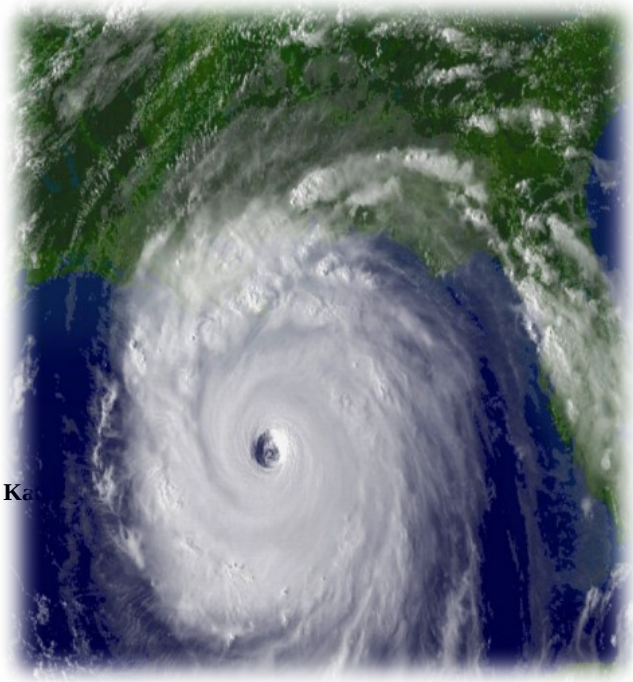
- What deliberations do attorneys who practice criminal law have to make when defending their clients?
- Which questions should they ask and of whom to persuade the judge or the jury? (Consider how detectives solve crimes.)
- How do they examine evidence and infer conclusions from a crime scene?
- What considerations must civil engineers make before designing a bridge or researchers prior to developing a new drug?
- What about city planners designing roads to relieve traffic congestion? All of these individuals share a common trait.

To be successful, they must be able to confront complex problems, discriminate between relevant and irrelevant information, and create inventive solutions. They must also be able to devise reasonable alternatives to failed decisions, communicate problems and ideas accurately to other people, and predict the consequences of bad judgment calls.

The previous example involving the radio ad was fairly straightforward. Thinking through the consequences of going into considerable debt for a quick $700 cash for most of us is a "no brainer." Consider, though, a real-life drama where your decisions and critical thinking mean life-and-death consequences for others. For example, place yourself in the role of an administrator whose hospital is located on the Gulf of Mexico. A Category Five hurricane, the highest on the Saffir-Simpson scale, is predicted to strike within 24 hours.

Life is the sum of all your choices.—
Albert Camus

The hurricane is named Katrina; the city is New Orleans —what do you do? Do you evacuate over 700 patients, many are elderly, frail and on life-support systems? Can these critically ill patients even be moved; and if so, where will they go? Or, should everyone ride it out hoping the hospital can withstand winds over 145 miles per hour? What if the levees break and the Mississippi River floods the hospital's emergency generators? How can the hospital serve the injured after the hurricane? What staff needs to remain on duty? How do you keep the lines of communication open? Obviously, you must make many decisions quickly, solve problems before, during, and after the hurricane, anticipate and deal with many crises as they emerge. Ultimately the final question is this: have you developed the necessary critical thinking skills to analyze the problem, examine alternatives, create solutions and live with your decision?



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Although the above may seem an extreme example, everyone in the path of Hurricane Katrina—from the politicians to school and hospital administrators, from emergency responders to thousands of businesses and the National Guard—had to plan how they would respond to the hurricane's imminent threat. Residents were forced to make life-and-death decisions such as leaving the city or staying put, riding it out in a shelter or at home. Some decided to evacuate the city while others, faced with rapidly rising water, hunkered in their attics or swam onto their rooftops. Regardless of who they were, all had one thing in common—the decisions they made and their choice of actions (or lack of actions) rendered significant consequences.

Having read these examples calling for critical thinking, are you ready to develop a definition yet or at least compose a list of desirable characteristics describing thinking critically? Take a minute now to jot down your ideas and answer the question: *Just what is critical thinking*? As you continue to *think about thinking*, (but before we suggest some helpful definitions), we must first explore the characteristics and benefits of critical thinking.

Characteristics of Critical Thinkers

If you were to list the characteristics of individuals you believe possess the qualities of a critical thinker, what adjectives would you list? Although not exhaustive, we have complied below our own list of unique characteristics that distinguish critical thinkers. How many of these characteristics apply to you?

- **curious** by desiring to investigate world-wide issues,
- **flexible** by possessing a willingness to consider other points of views,
- **rational** by thinking logically and avoiding fallacious reasoning,
- **watchful** for new opportunities to intellectually develop,
- **creative** by exploring unconventional approaches to ideas and problems,
- **self-directed** by demonstrating leadership when solving problems,
- **truthful** by assessing one's own prejudices,
- **self-confident** by building on previous mistakes and assessing one's own abilities,
- **aware** by appreciating the complexities of the human condition,
- **reflective** by displaying a wiliness to assess one's own limitations.

Benefits of Critical Thinking

Who among your friends, professors, family, or work associates are good decision makers and problem solvers? Are these individuals admired for their ability to make reasoned judgments and clarify complex issues? Do these individuals communicate their ideas better than others? Are they capable of interpreting and making effective assessments of problems? Can they generate innovative ideas? Are they capable of keeping a clear head when others become frantic? Do they exhibit common sense? Do they seem to possess thinking skills others have not acquired?

Thinking is a mental process; improving our ability to think promotes our well-being and allows us to maximize our potential. The people around us who excel in their professions have one thing in common: they desire to excel and maximize their potential. Being able to think critically, then, directly affects our ability to excel and transform the quality of our lives. Everything from deciding what we will eat for supper tonight to determining our career paths is influenced by our previous experiences and knowledge. Being able to make reasoned, informed decisions based on these experiences and knowledge is, well, priceless.

Therefore, critical thinking can be seen as a disciplined, intellectual process that requires reasoned judgments. Being able to think deliberately and decisively allows us to make better informed decisions as well as comprehend how others' decisions may impact us. Someone who has developed the skills for critical thinking is rational and keeps a clear head when others may panic. Critical thinkers maintain a vigorous degree of skepticism and do not easily fall for scams or tricks. Furthermore, they work smarter, often are the most successful in their careers and possess general feelings of self-confidence and self-awareness.

Choices are the hinges of destiny—Pythagoras

Definition: By now you should have developed your own working definition for *critical thinking*. Perhaps you came up with something similar to Richard Paul, a reformer in the critical thinking movement, who suggests critical thinking is really "thinking about thinking in order to

improve your thinking." Or, perhaps you came up with a more formal definition similar to one of the following:

- "Critical thinking is the intellectually disciplined process of actively and skillfully conceptualizing, applying, analyzing, synthesizing, and/or evaluating information gathered from, or generated by, observation, experience, reflection, reasoning, or communication, as a guide to belief and action." (Paul Scriven)

- "Most formal definitions characterize critical thinking as the intentional application of rational, higher order thinking skills, such as analysis, synthesis, problem recognition and problem solving, inference, and evaluation." (T. A. Angelo)

- "Critical thinking is the ability to think about one's thinking in such a way as 1. To recognize its strengths and weaknesses and, as a result, 2. To recast the thinking in improved form." (Center for Critical Thinking)

- "Critical thinking... means making reasoned judgments." (B. K. Beyer).

Barriers to Critical Thinking

By now you have a good overview of what is meant by *critical thinking*. In the next two chapters we open up the CT/CW (critical thinking/critical writing) toolkit further by introducing you to universal standards for critical thinking as well as problem solving strategies. Then, in subsequent chapters we explore the principal facets of cognition and show you how to significantly improve the quality of your thinking and writing. However, first, we ask you to pause briefly to consider the most important aspect affecting your overall success—your motivation toward learning.

The following list contains the most common barriers students report for lacking the motivation to learn. Use the following questions as a sample guide to evaluate your own attitude toward learning. Which, if any, of the following apply to you?

Category I

- I'm not receptive to new ideas.
- I'm not interested in my classes.
- I'm not confident in my abilities to reason through a problem.
- I don't feel I'm very creative or have a good imagination.
- I feel others may think I am incompetent.
- I get defensive when questioned about my way of thinking.
- Often I fear that I am wrong.
- I'm intimidated by others who seem smarter than I am.
- I was never good at writing.
- I have a poor educational background.

Or, perhaps your attitude is best reflected by the following student responses:

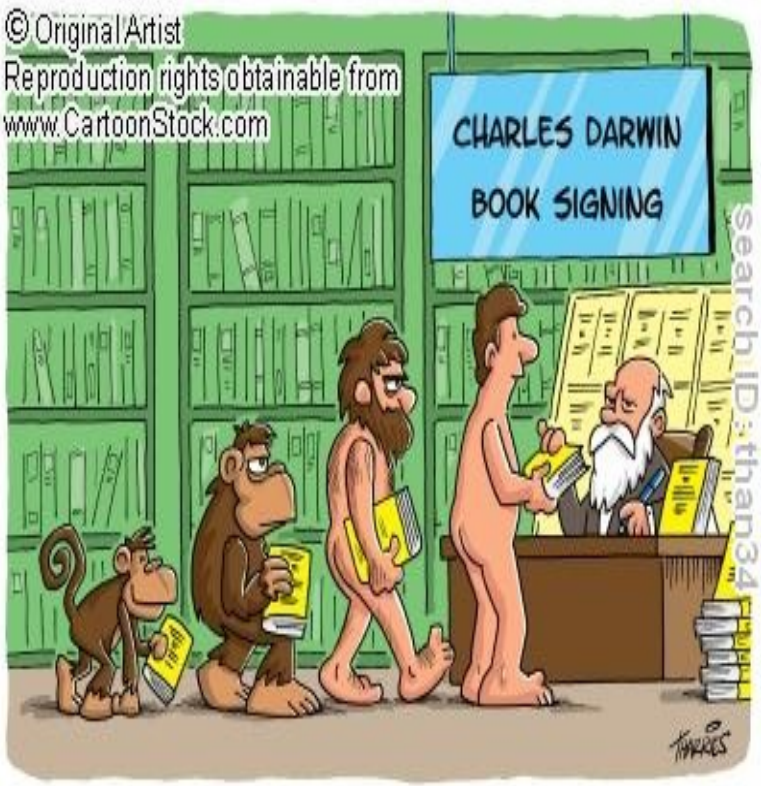
Category II

- I approach new learning with a positive attitude.
- I'm motivated to learn because I understand how education will affect my future.
- I don't mind committing the necessary time and energy to learning.
- I am naturally curious and want to learn as much as possible.
- I find problem solving challenging.
- I like writing.

What can individuals do if they fall into the first category? Can they become more motivated and develop more self-confidence? Research indicates that the lack of motivation is often the result of individuals lacking confidence in their intellectual ability which, in turn, creates anxiety of future success. Although *Critical Thinking/Critical Writing* is not intended specifically as a "self-help" book, we strongly believe critical thinking skills can be acquired with the right attitude and discipline. Becoming a good critical thinker and learning to write with purpose and clarity will certainly enhance if not completely change an individual's attitude toward learning. It means gaining new confidence and will lead to future opportunities.

THE RELATIONSHIP BETWEEN CRITICAL THINKING AND CRITICAL WRITING

Charles Darwin called his process ″Brain Rioting.″ After returning from his famous voyage to the Galapagos Islands, Darwin relates in his journal he was perplexed by the numerous variations he found within the same species of animals. Why, he questioned, do the finches that live on one island have longer and stouter bills than the finches on the other islands? Why do the tortoises that inhabit the different islands have differently shaped shells? To answer these and many other enigmatic questions, Darwin turned to "brain rioting," a deliberate and concerted attempt to analyze these complex questions. Sitting at his desk for hours on end, Darwin "brain rioted" by rapidly scrawling down ideas, questions, and conclusions as they came to him. Writing down his observations from the Galapagos Islands prompted new ideas to emerge while requiring others to be discarded. Gradually, what were once disorganized, random thoughts began to connect and unify on paper. Finally, after many revisions and alterations, Darwin published *The Origin of Species* and the theory of natural selection was born.



For Darwin, then, *writing* down his thoughts caused him to clarify and assimilate his thoughts into a calculated, reasoned approach to solving a problem. Ask yourself, "How many times did I discover what I really thought about an issue only *after* I began writing about it?" Writing is a *focused* form of thinking. It is a process of discovery where the writer unearths what he or she really thinks to be true. Many confuse writing as a physical process of something to do after thinking through an idea. Instead, research shows that writing begins as a private process of self-discovery that evolves into public discourse through many iterations and revisions. Writing brings sense to our thinking by concentrating our thoughts from randomness into intelligible, meaningful ideas we can share with others.

This, then, is the purpose of this text —to demonstrate the relationship between critical thinking and critical writing, to provide you with the necessary tools and skills to make you a better critical thinker and writer, and to help you develop your academic voice. Writing will be used as the medium to accomplish these objectives and discipline your thinking. As you work through the subsequent chapters, you will gain confidence in your ability to think and write critically. In addition, you will greatly improve your analytical skills and learn how to evaluate the quality of others' thinking as well as your own.

Implications

In October, 1957 the Soviet Union launched the first satellite called Sputnik. This singular event significantly accelerated the world's quest for knowledge. Fearing the Soviets would eclipse it as a superpower, the United States government initiated hundreds of new research and development projects to catch up with the Soviets' apparent lead in science. U. S. schools immediately recalibrated their curriculums to include more emphasis on math, science, and engineering. The cascading effects from Sputnik became the genesis for the Knowledge Explosion, the seminal point in history where knowledge of ourselves and our universe began growing exponentially. Professor Solomon Diamond writes in his book *The World of Probability* that we know "about two million times as much about the universe we live in and the creatures which inhabit it as was known shortly after the death of Galileo." In 1971 the *Christian Science Monitor* reported "the world's population is doubling every 35 years,

Nothing in all the world is more dangerous than sincere ignorance and conscientious stupidity.—
Martin Luther King, Jr.

Registered Nurse Critical Care

Work Location: United States-Florida-Miami-Kendall Regional Medical Center - Kendall

Description

Registered Nurse Critical Care

The ICU/CCU RN manages the nursing care of critically ill patients in accordance with established policies, procedures and protocols of the healthcare organization. The tasks and responsibilities include:

- Assesses plans and evaluates patient care needs.
- Carries out physician orders.
- Administers prescribed medications, changes dressings, cleans wounds, monitors vital signs.
- Provides respiratory and ventilator management and monitoring for patients.
- Serves as the primary coordinator of all disciplines for well coordinated patient care.
- Monitors, records and communicates patient condition as appropriate utilizing computerized documentation systems.
- Instructs and educates patients and families.
- Assesses and coordinates patient's discharge planning needs with members of the healthcare team.
- Provides age and culturally appropriate care.
- Orients and mentors new staff members.
- Follows Standard Precautions using personal protective equipment as required.

Required Job Qualifications include:

- Current licensure as an RN.
- Current certification as a BLS Healthcare Provider and ACLS.
- Graduate of an accredited school of professional nursing.
- Minimum of one year of experience in critical care in a hospital setting
- Customer service abilities including effective listening skills.
- ***Critical thinking skills, decisive judgment and the ability to work with minimal supervision in a fast paced environment.***

but the accumulation of scientific data, which a century ago was doubling every 100 years, now is doubling every 6 years!" Most recently, by the end of 2012, many theorists believe our store of human knowledge will double every two or three years. Dr. J. E. Wallen of the Smithsonian Institute summarizes this dilemma: "I spend 25 percent to 50 percent of my working time trying to keep up with what's going on."

So, within little over a decade from that first Sputnik flight in 1957, the U.S. landed men on the moon, and while still in their infancy, computers, the internet, cell phones, video cameras, and thousands of new technological products and advances emerged radically reshaping the world. Thomas Friedman argues in his bestseller *The World is Flat* that as a result of this continuing, relentless pursuit of knowledge, that in just the first decade of the 21st century the world, international boundaries have diminished significantly "flattening" the world. Historians, he predicts, will eventually view 2000-2010 as an unprecedented watershed where the role of governments, business, education, research, science, politics but particularly the individual will be transformed immeasurably.

What, then, are the implications for students furthering their education and considering their life's work? If knowledge actually does double every few years, how can anyone become competent, much less stay current in any given field? You probably have already guessed the answer. Many careers today require advanced degrees or employees to continually update their skills by returning to school or special classes. Yet, additional education alone is not sufficient. Without the critical thinking and critical writing skills to discipline the new learning, learning by itself would be a waste of time.

Take a moment to read the adjoining job ad for a Registered Nurse Critical Care. Notice that among the many duties, the applicant's qualifications must include being able to *assess, evaluate, instruct, communicate,* and *coordinate* —all skills requisite to critical thinking. Even the final qualification listed in the ad specifically

requires the applicant to possess *critical thinking skills, decisive judgment and the ability to work with minimal supervision in a fast-paced environment.*

The explosion of knowledge, then, with the changes, complexities, and challenges it has created, calls for a revolution in the way we think and approach problems. Much of what we learn today will be obsolete in a few years. Change has become permanent. A new breed of worker is emerging, one who must take over the tasks once performed by many workers, one who must be able to make good judgments and solve problems. Many will work in teams; many will work from home. Successful individuals will rise to this challenge. The late Steve Jobs, founder of Apple, Inc., told his employees, "I have a great respect for incremental improvement, and I've done that sort of thing in my life, but I've always been attracted to the more revolutionary changes. I don't know why. Because they're harder. They're much more stressful emotionally. And you usually go through a period where everybody tells you that you've completely failed."